

Rajeev Mutyalu

Founder, Learn with Charlie • Lead Software Architect • Production Technology Architect • **Engineering & R&D Leader**

Founder, Learn with Charlie • Selected Production Technology Credits • 1917 • RRR

Former Technicolor R&D Supervisor • Technicolor Fellowship Network Associate Member

UK Skilled Worker Visa • Existing Sponsorship in Place • Available Immediately

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🎯 Executive Summary

Founder of Learn with Charlie and former Technicolor R&D Supervisor with 20+ years of experience across VFX production technology, software architecture, production engineering and global engineering leadership. Founder of **Learn with Charlie**, a curriculum-bound AI maths tutor for Indian school children that teaches from their own textbook with natural voice, live blackboard, and zero guessing. Proven track record designing and evolving production-critical platforms, proprietary applications and DCC-integrated systems across Astra Studios, Technicolor Group and MPC Film spanning OpenUSD architecture, global review platforms, modular Nuke ecosystems, Editorial Turnover / Conform / I/O systems, and studio automation.

Led distributed engineering teams across London, Montreal and Bengaluru, defining engineering standards and translating production requirements into scalable, maintainable technical systems across major productions including *1917* (Academy Award for Best Visual Effects), *RRR*, *Mufasa: The Lion King*, *Back in Action*, *Spaceman*, and *Prehistoric Planet*. Associate Member of the Technicolor Fellowship Network Committee and recipient of 9+ studio leadership & excellence awards.

⚡ Architecture & Technical Competencies

Platform Architecture

🏛️ Product & Platform Architecture

- **Production Technology Architecture:** Production-critical applications, cross-DCC systems, platform integration, production workflow architecture and engineering standards.
- **Platform Evolution:** Productionising R&D technologies, aligning cross-site technical architecture, integrating application ecosystems and standardising studio workflows.

Production Systems

🎥 VFX Production Systems & DCC APIs

- **DCC APIs:** Nuke C++/Python API, Maya Python API, Hiero / Nuke Studio Python API, 3D Equalizer, Silhouette.
- **Production Tracking:** ShotGrid / Shotgun Python API, webhooks, metadata automation and task/status integration.

Engineering Leadership

👥 Engineering Leadership & Delivery

- **Distributed Teams:** Leading R&D Software Engineering, Development and TD teams across London, Montreal and Bengaluru.
- **Engineering Standards:** Architectural stewardship, engineering practices, Agile/Scrum delivery, technical mentorship and alignment between creative supervisors and engineering teams.

OpenUSD Architecture

📦 OpenUSD Production Architecture

- **USD Architecture:** OpenUSD (USD Python API, Composition Arcs, Layer Stacking, Edit Targets, Sublayering), asset & shot schemas for full-CG and live-action VFX.
- **Colour & Timeline Standards:** OpenColorIO (OCIO), ACES Colour Pipeline, OpenTimelineIO (OTIO) sequence interchange.

Core Systems

🌀 Core Software & Systems Architecture

- **Languages & UI:** Python 3.x (Expert), PyQt / PySide (UI/UX Engineering), C++, Java, Objective-C, Cocoa, Shell Scripting.
- **Infrastructure:** Docker, Linux/Unix environments, REST APIs, FastAPI, Git, Tractor Render Farm management & dispatching.

Conform & I/O

🎞️ Editorial Turnover, Conform & I/O

- **Turnover Ingest:** Package inspection, automated DI discrepancy reporting, regex pattern generation, multi-track Hiero / Nuke Studio conform.
- **Colour & Plates:** CDL grade extraction, shot LUT application, frame range validation, automated USD plate sublayering.

Standardisation

🌐 Production Integration & Cross-Studio Standardisation

- **Multi-Facility Toolsets:** RPM Unification across MPC, MR.X, and The Mill, standardising compositing and tracking toolsets.
- **Shared Application Frameworks:** Reusable PyQt components, cross-studio software distribution and consistent operational workflows.

EdTech & AI Platforms

🚀 EdTech & AI Platforms (Learn with Charlie)

- **Learn with Charlie:** Founder of Learn with Charlie (learnwithcharlie.ai), curriculum-bound AI maths tutor for Indian school children; Studio.AI generative video platform with ComfyUI node workflows and ACES colour conforming.
- **Verified Architecture & Governance:** Deterministic SymPy engine (zero guessing), ElevenLabs voice, multi-agent AI teams under 50 ADRs, and n8n zero-touch automation hubs.

MCP & Agentic

🕯️ MCP / Agentic Systems

- **Model Context Protocol (MCP):** Custom MCP server engineering, JSON-RPC 2.0 protocol, tool & resource schemas, pipeline API binding.
- **Agentic Tooling:** Claude Code (CLI), OpenAI Codex, Google Antigravity, subagent orchestration, task decomposition.

Private AI

🔒 Private / On-Premise AI

- **Local Inference:** Ollama, air-gapped studio deployments, local GGUF / AWQ model quantisation, context window optimisation.
- **IP Protection:** Secure internal AI endpoints designed to keep sensitive studio data within controlled infrastructure and reduce exposure to public AI services.

💼 Professional Experience

FOUNDER • AI EDTECH PRODUCT

Aug 2026 – Present

Founder, Learn with Charlie

Charlie AI • London, UK

Focus: Platform Architecture, Offline PyMuPDF Pipeline & Engineering Governance

- **End-to-End Platform Architecture:** Architected and engineered a resilient multi-tier platform comprising React & TypeScript (Vite), FastAPI (Python 3.12), PostgreSQL, Docker Compose, Caddy reverse proxy, and Cloudflare Workers edge delivery.
- **Offline Content Pipeline:** Designed an automated ingestion and compilation pipeline using PyMuPDF and SymPy, turning school textbook chapters into structured lesson scripts, parameterised templates, and verified exercise checks under human-approved review workflows.
- **Engineering Governance & Agent Direction:** Directed a team of specialised AI coding agents (frontend, backend, content pipeline, independent test and maths reviewers, plus 3 automated student testers) under 50 ADRs, one concern per pull request (115+ PRs), and owner-approved merges.
- **Deterministic Symbolic Maths Engine:** Integrated a SymPy symbolic engine ensuring 100% mathematical verification for every step and line of working, preventing model hallucinations and giving line-by-line error feedback without giving answers away.
- **Privacy by Design & Global Readiness:** Implemented privacy by design with pseudonymous IDs and minimal student data, full locale-keyed text internationalisation, and business outreach to Indian textbook publishers.

CURRENT LEADERSHIP ROLE

Sep 2025 – Present

Lead Software Architect

Astra Studios • London, UK

Focus: Production Technology, OpenUSD, VFX Platforms & AI Systems

- **Engineering & Technical Leadership:** Lead Development and TD teams, defining production technology architecture and engineering standards while working directly with artists, production, supervisors and leadership to translate creative requirements into scalable technical systems.
- **Production Platform & OpenUSD Architecture:** Architected an end-to-end production-grade OpenUSD architecture connecting asset and shot workflows across DCCs and departments for full-CG and live-action VFX.
- **Editorial Turnover, Conform & I/O:** Engineered the studio's Editorial Turnover, Conform & I/O pipeline from scratch, integrating package inspection, DI discrepancy reporting, regex pattern generation, multi-track Hiero conform, CDL grades, shot LUTs and USD plate sublayering.
- **Studio.AI GenAI Platform & Workflows:** Integrated generative AI workflows into studio infrastructure, including Studio.AI prompt-to-pipeline automation, ComfyUI node workflows, and artist-facing controls with ACES colour conforming.
- **Zero-Touch Studio Automation:** Orchestrated zero-touch n8n studio automation hubs using inbound and reverse webhooks for automated show onboarding, smart plate validation, and ShotGrid metadata injection.

Mar 2023 – Feb 2025

R&D Supervisor

Technicolor Group • London, UK

- **Global Review Platform Architecture:** Spearheaded the architecture and productionisation of ReviewTool, a multi-facility VFX review platform with real-time 4K+ media caching, synchronisation and RV integration supporting distributed production workflows across global studios.
- **ProNuke Application Ecosystem:** Architected and engineered a modular Nuke production ecosystem spanning automated shot assembly, sequence-level publishing and batch render orchestration across global artist teams.
- **Meshroom Photogrammetry in Production:** Productionised Meshroom into matchmove and tracking workflows, cutting camera extraction turnaround from days to hours with fast switching between Meshroom, Maya, and 3D Equalizer.
- **Global R&D Leadership:** Led globally distributed R&D software engineering teams across London, Montreal, and Bengaluru supporting Compositing, Roto, Paint Prep, Camera Tracking, and Rotoanim departments.
- **Conform Ingest Automation:** Supervised Conform Ingest workflows using Hiero and Nuke Studio, automated CDL and LUT grade processing, and ShotGrid frame range validation.

Jan 2022 – Feb 2023

R&D Supervisor

Technicolor Group • Bengaluru, India

- **2D Software Engineering:** Headed 2D software engineering and pipeline development across Bengaluru and Montreal hubs.
- **Review Platform Prototyping:** Architected core compositing automation tools and early prototypes of the studio VFX review platform (ReviewTool).
- **Agile Execution:** Mentored engineers, established Agile and Scrum sprint frameworks, and aligned dev priorities with production executives.

Jul 2017 – Dec 2021

Lead Pipeline Software Developer

MPC Film • Bengaluru, India

- **Feature Production Pipelines:** Led core 2D pipeline solutions across Academy Award-winning films and global tentpole productions (1917, RRR, The Lion King).
- **Cross-Studio Platform Standardisation:** Spearheaded RPM Unification across MPC, MR.X and The Mill, consolidating and standardising software toolsets and UI/UX across compositing and tracking workflows to improve technical consistency across multiple studio environments.
- **Nuke Bridge Hub:** Developed Nuke Bridge integrating Tessa (AMS), ShotGrid (PMS), and Tractor (Render Farm), establishing a version-controlled, high-throughput pipeline.
- **Render Dispatcher:** Engineered the Nuke Render Dispatcher with automated render order traversal and Reserve & Commit dispatch mechanics.

Jul 2012 – Jun 2017

Team Lead – Technology & R&D Production

Technicolor • Bengaluru, India

- **Media Engines:** Architected macOS AVFoundation and QuickTime media processing engines (Channel Mapper, Frame Rate Converter, Compliance Edit Lists).
- **Team Management:** Managed an engineering team of 7 developers across C++, Objective-C, Cocoa, and Java, optimising production compression and authoring pipelines.
- **Hardware Compliance:** Served as primary technical contact for player manufacturers, resolving Blu-ray authoring framework and playback compatibility issues.

May 2008 – Jun 2012

Senior Software Engineer – Technology

Technicolor • Bengaluru, India • Los Angeles, USA • Hannover, Germany

- **Disney Collaboration (Los Angeles):** Collaborated on-site with Disney engineering in Los Angeles to architect the Fast Menu Load (FML) template and integrate Generation 2 Bootloaders (G2B).
- **World First 3D Blu-ray (Hannover):** Engineered core 3D capabilities into the BD-J Authoring framework at Technicolor R&I in Hannover, Germany, as a key contributor to the world's first released 3D Blu-ray disc.
- **Founding Engineer:** Founding software engineer at Technicolor India, building runtime alignment engines ('Graphics Adjuster') and studio authoring templates.

Sep 2006 – Apr 2008

Application Engineer & Game Programmer

e4e & Dhruva Interactive • Bengaluru, India

- Ported mobile game titles to Nokia, Motorola, Samsung, and Sony Ericsson devices for T-Mobile and Verizon; engineered touchscreen interfaces for early mobile OS platforms.

Honours, Awards & Fellowship

Technicolor Fellowship Network

2015 - 2025

Associate Member • Technicolor Group

> *"Awarded for production environment expertise, strong customer relationships, and deep collaboration with Research & Innovation—delivering a rare mix of creativity, pragmatism, and architectural stewardship."*
>
> — Fellowship Nomination Committee • Mar 2015

Fellowship Committee Election R&I Collaboration

Architectural Leadership

Studio Leadership & Excellence

2008 - 2024

9+ Honours • Technicolor, MPC Film & e4e

- **Value Champions & Best Employee of the Month (Aug 2019)** — *Visionary, Focused, Quietly Brilliant (Technicolor FEV)*
- **Performer of the Quarter • Team Lead (Oct 2016)** — *Technology R&D leadership*
- **Performer of the Quarter (2011, 2012, 2014)** — *Disney collaboration & pipeline innovation*
- **Extra Miler & Executive Team Awards (2011, 2013)** — *DVD/Blu-ray software & dedication*

Recent AI Professional Development

AI Generalist Accelerator Program

Generative AI Business Applications • Outskill

View Certificate ↗

> *14-Day Intensive Generative AI Accelerator mastering practical automation, MCP server tool deployment, AI agent systems, and no-code product development.*
>
> — Outskill • Aug 2026

Prompt Engineering n8n Workflow Automation

MCP Tool Integration Voice Agent Orchestration

Multimodal AI & Cloning

AI Accelerator C13 Hackathon

View Award ↗

2-Day Innovation Challenge • Outskill

> *Awarded for rapid prototyping, creativity, and deployment of functional generative AI application architectures during the 2-Day AI Innovation Challenge.*
>
> — Outskill Hackathon Jury • Aug 2026

Rapid AI Prototyping Agent System Architecture

Live Product Deployment

Education & Academic Background

Bachelor of Technology (B.Tech) • Computer Science & Engineering

2002 - 2006

Jawaharlal Nehru Technological University (JNTU) • Sri Venkateswara College of Engineering & Technology (SVCET)